Kermina David

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Education

New Jersey Institute of Technology - Albert Dorman Honors College Newark, NJ

Bachelor of Science in Electrical Engineering (NJIT Honors - Full Scholarship) *Sept. 2023 – May 2027*

- Minor in Grand Challenges: Interdisciplinary program focused on solving global engineering challenges through innovative hardware and software solutions

Experience

Software Engineering Intern June 2026 – August 2026

Prudential Financial - Process Excellence Team *Newark, NJ*

- Served in a project-management role, driving evaluation and onboarding of new technologies through workflow documents and Process Design Documents (PDDs)
- Built an AWS-based audio transcription and note-taking tool that generates workflow documents and PDDs directly from recorded meetings, saving ∼30 hours of manual documentation effort
- Led hackathon team to build a learning platform that makes AI accessible and teaches effective prompting; currently being deployed by PGIM

Software Engineering Intern June 2025 – August 2025

Prudential Financial - Infrastructure Team *Newark, NJ*

- Developed DNS health monitoring application using FastAPI and Docker for real-time infrastructure monitoring across enterprise network
- Architected HashiCorp Vault integration template for secure credential management across 60+ service accounts
- Led hackathon team to develop parking and desk booking system, integrating IoT sensors with backend API infrastructure

Biomedical Research Assistant April 2024 – Present

NJIT BME Department/Rutgers SHP *Newark, NJ*

- Designed and developed gamified therapy videogame using hand and voice control interfaces for cerebral palsy rehabilitation
- Integrated hardware sensors and signal processing algorithms to create 8-week therapeutic gaming protocol, achieving 60% improvement in patient therapy compliance and motor function progression

Projects

Verdura Automations (Co-Founder) | *ESP32, Raspberry Pi, Node.js, React/TypeScript* Sept. 2025 – Present

- Co-founded IoT greenhouse automation startup and raised $30K in funding to build and deploy the platform
- Designed battery-powered ESP32 sensor nodes integrating soil moisture (VH400), soil temperature (THERM200, DS18B20), humidity (VG-HUMID), and light (LT150) sensors, with MAX17048 fuel-gauge power monitoring and EMC2101 fan control
- Designing custom PCB integrating the ESP32, sensor interfaces, and power management circuitry for a compact, deployable node board
- Built Raspberry Pi Node.js/Express/SQLite backend with mDNS device discovery and a React/TypeScript dashboard; led a four-person mobile app team and cost-modeled AWS vs. Supabase for scale

Technical Skills

Languages: C/C++, Python, Java, JavaScript/TypeScript (Node.js, React), C#, MATLAB, SQL
Cloud & DevOps: AWS, Docker, Kubernetes, Git, Jenkins, CI/CD, HashiCorp Vault, microservices
Hardware & Embedded: ESP32, Arduino, Raspberry Pi, sensor integration, PCB design, SolidWorks
Tools & Libraries: Unity, NumPy, SciPy, OpenCV, TensorFlow, Multisim, VS Code

Honors & Certifications

Honors: IEEE AP-S Poster Competition Winner, Dana Knox Showcase - Top 20 (*April 2024*)
Certifications: AWS Cloud Practitioner (*August 2025*)
Languages: English (Native), Arabic (Fluent), Spanish (Conversational), Japanese (Basic)